On Different Flavors hence Needs for *Tabular Reasoning*

Madelon Hulsebos

27 October 2026
Booking.com
Amsterdam

CWI TRL Lab ellis UNIT AMSTERDAM

Madelon Hulsebos

Academic trail

BSc/MSc — TU Delft

PhD (external) — University of Amsterdam

CWI

ellis UNIT AMSTERDAM

PRIOR

Research student — MIT

Postdoc — Berkeley, University of California

Industry

- Data Scientist, 2019–2021 (NL)
- Industry Researcher 2021–2022 (US)



madelonhulsebos.com
trl-lab.github.io

Current:

- **Tenured Researcher** at CWI, leading the Tabular Reasoning & Learning (TRL) Lab*
- **Faculty & MT Member** ELLIS unit Amsterdam
- **Scientific Advisor** Prior Labs (🇩🇪Tabular AI startup)

*founded TRL workshop series @neurips/acl in ‘22

TRL Lab @ CWI

Research vision: prevailing actionable critical

Democratizing insights** from **tabular data

1) **DS/DA from 1 to 10**; so data scientists can focus on things that matter
2) **everyone else from 0 to 1**; democratize insights from data

TRL Lab



Agentic systems for relationship discovery



Tabular insight generation

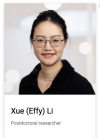


Multi-modal tabular learning (predictive)

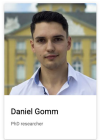


Con*text*-rich (pred) tabular learning

incoming



Knowledge-grounded agentic data science



End-to-end tabular QA

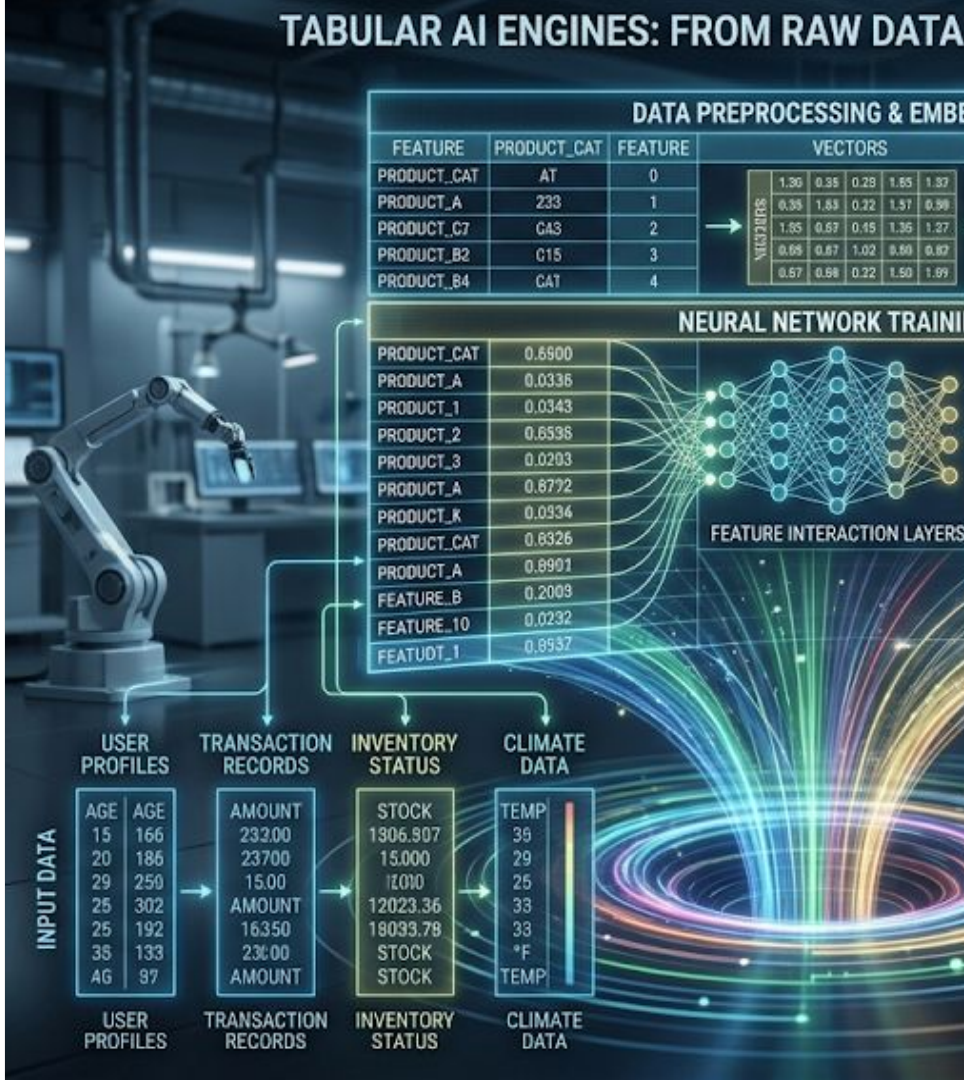




Scaling tabular foundation models

https://trl-lab.github.io/

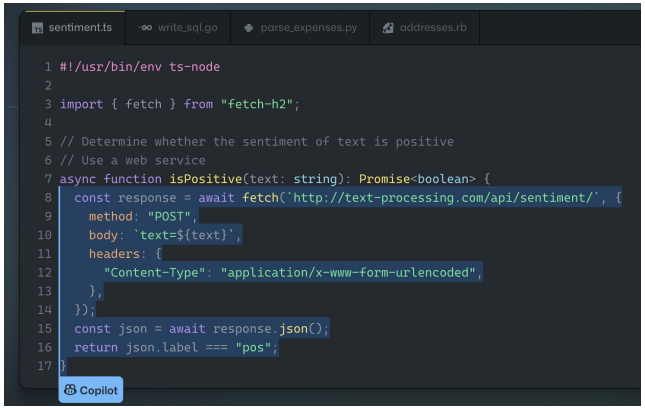
The case for Tabular AI



AI has made big leaps in…

Lorem ipsum dolor sit amet, consecte
adipiscing elit. Maecenas porttitor co
Fusce posuere, magna sed pulvinar u
lectus malesuada libero, sit amet com
eros quis urna. Nunc viverra imperdi
est. Vivamus a tellus. Pellentesque ha
tristique senectus et netus et malesua
turpis egestas. Proin pharetra nonum
Mauris et orci. Aenean nec lorem. In
Donec laoreet nonummy augue. Susp





Text

Images

Code…

crop rotation : Tabelle

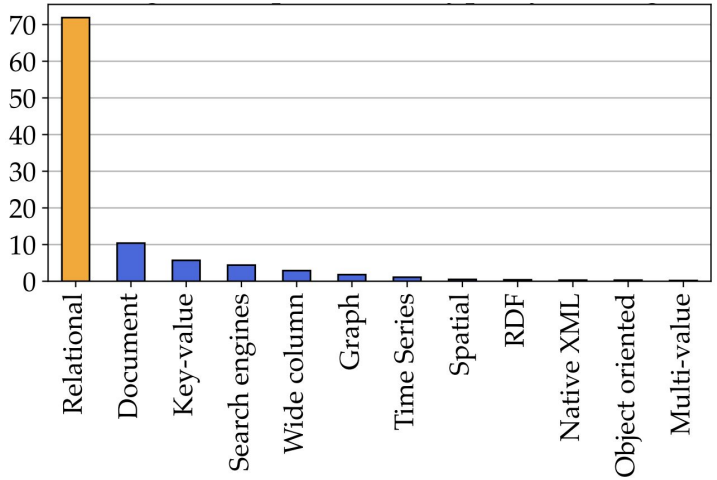
Nr	ID	seed rate	yield	crop	cultivar	pre crop	pre-pre crop	pre-pre-pre	soil type	precipita	tempera	comment
1	68		91	winter wheat		sugar beets	beans		sandy loam, loe	636	9,6	wb, sg,
2	68		100	winter wheat		sugar beets	rotation fallow		sandy loam, loe	636	9,6	cultivation
3	68		97	winter wheat		sugar beets	fallow land (5,5y)		sandy loam, loe	636	9,6	1993-1996
4	136		95	winter wheat		oats	sugar beets		sandy loam, loe	636	9,6	
5	136		96	winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	cultivation
6	136		107	winter wheat		sugar beets	maize		sandy loam, loe	636	9,5	1991-1994
7	136		107	winter wheat		sugar beetsn	summer wheat	maize	sandy loam, loe	636	9,5	
8	136		82	winter wheat		oats	sugar beets	sugar beets	sandy loam, loe	636	9,5	organic
9	136		77	winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	organic
10	136		85	winter wheat		sugar beets	maize	maize	sandy loam, loe	636	9,5	organic
11	136		84	winter wheat		sugar beets	summer wheat	sugar beets	sandy loam, loe	636	9,5	organic
12	57	371	98	winter wheat	Sperber	sugar beets	winter barley	winter wheat	sandy loam, loe	635		wb, ww
13	57	365	98	winter wheat	Sperber	potatos	sugar beets	summer barl	sandy loam, loe	635		cultivation, weed
14	57	365	105	winter wheat	Sperber	sugar beets	maize	maize	sandy loam, loe	635		1987-1992
15	57	365	97	winter wheat	Sperber	sugar beets	winter wheat	sugar beets	sandy loam, loe	635		
16	39	433	90	winter wheat	Okapi	summer barley			sandy loam, loe	690	8,5	oats, cultivation, weec
17	39	433	100	winter wheat	Okapi	oats			clay, silt	690	8,5	1982-1986
18	39	433	97	winter wheat	Okapi	winter wheat			clay, silt	690	8,5	

Tables?

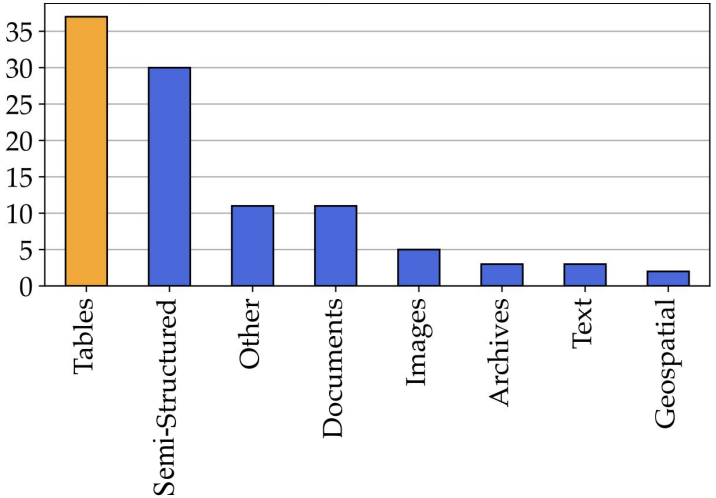
Why?

1 tables **prevail** in organizations

Popularity of database systems



Distribution Google Dataset Search



2 tables serve **high value** use-cases

Patient ID	Age range	MS age onset	OB	Pregnancy	Lesional load (YYYY/MM–number)	Spinal lesions (YYYY/MM–number)	Treatment
1070	20-29	20	yes	no	2007/03– 32007/06–5 2010/10· 2010/12·		2007/09: IFN-β1a
3736	30-39	30	yes	yes, from 2015/11 to 2016/08	2013/05· 2015/06· 2014/09· 2015/01·		
2789	30–39	33	yes	yes, from 2012/10 to 2013/07 and from 2015/11 to 2016/08	2013/11 2013/12 2014/09 2017/01·		

HEALTH

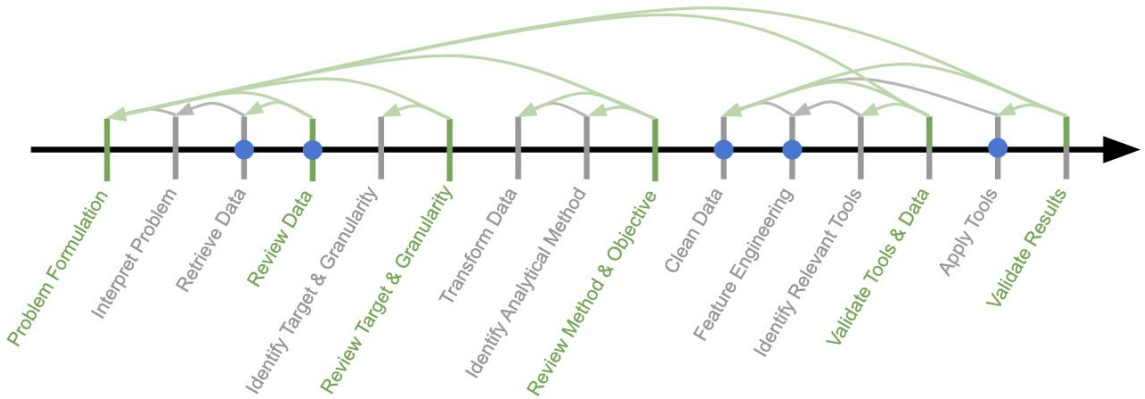
operation	admin0Name	admin0Pcode	numPresentIdpInd	reportingDate	displacementReason	numberMales	numberFemales	idpOriginAdmin1Name
Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR	193	2022-02-28T00:00:00	Conflict	67	126	Zaporizka
Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR	255	2022-02-28T00:00:00	Conflict	99	156	Odeska
Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR	2318	2022-02-28T00:00:0				
Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR	1054	2022-02-28T00:00:0				
Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR	651	2022-02-28T00:00:0				
Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR	336	2022-02-28T00:00:0				
Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR	2	2022-02-28T00:00:0				
Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR	246	2022-02-28T00:00:0				
Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR	190	2022-02-28T00:00:0				
Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR	6	2022-02-28T00:00:0				
Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR	436	2022-02-28T00:00:0				
Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR	8	2022-02-28T00:00:0				

PEACE

	A	B	C	D	E	F	G	H	I	J
1	© Corporate Finance Institute. All rights reserved.		Historical Period				Forecast Period			
2	**Financial Model**		**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**
60										
61	**Income Statement**									
62										
63	**Revenue**		**118,086**	**131,345**	**142,341**	**150,772**	**165,849**	**182,434**	**200,678**	**218,739**
64	Cost of Goods Sold (COGS)		48,004	49,123	52,654	56,710	69,657	78,447	88,298	98,432
65	**Gross Profit**		**70,082**	**82,222**	**89,687**	**94,062**	**96,193**	**103,987**	**112,379**	**120,306**
66	**Expenses**									
67	Marketing, Advertising & Promotior		22,658	23,872	23,002	25,245	28,194	31,014	34,115	37,186
68	General & Administrative		10,125	10,087	11,020	11,412	15,000	15,000	15,000	15,000
69	Depreciation & Amortization		18,150	17,205	16,544	16,080	7,504	9,003	10,203	11,162
70	Interest		2,500	1,500	1,500	1,500	3,000	3,000	1,000	1,000
71	**Total Expenses**		**53,433**	**52,664**	**52,066**	**54,237**	**53,699**	**58,017**	**60,318**	**64,348**
72	**Earnings Before Tax**		**16,649**	**29,558**	**37,622**	**39,825**	**42,494**	**45,970**	**52,062**	**55,958**
73										
74	Taxes		4,858	8,483	10,908	11,598	11,898	12,872	14,577	15,668
75	**Net Earnings**		**11,791**	**21,075**	**26,713**	**28,227**	**30,596**	**33,099**	**37,484**	**40,290**
76										

ECONOMY

3 **repetitive**, **multifaceted,** **multi-step** pipelines



potential to augment!

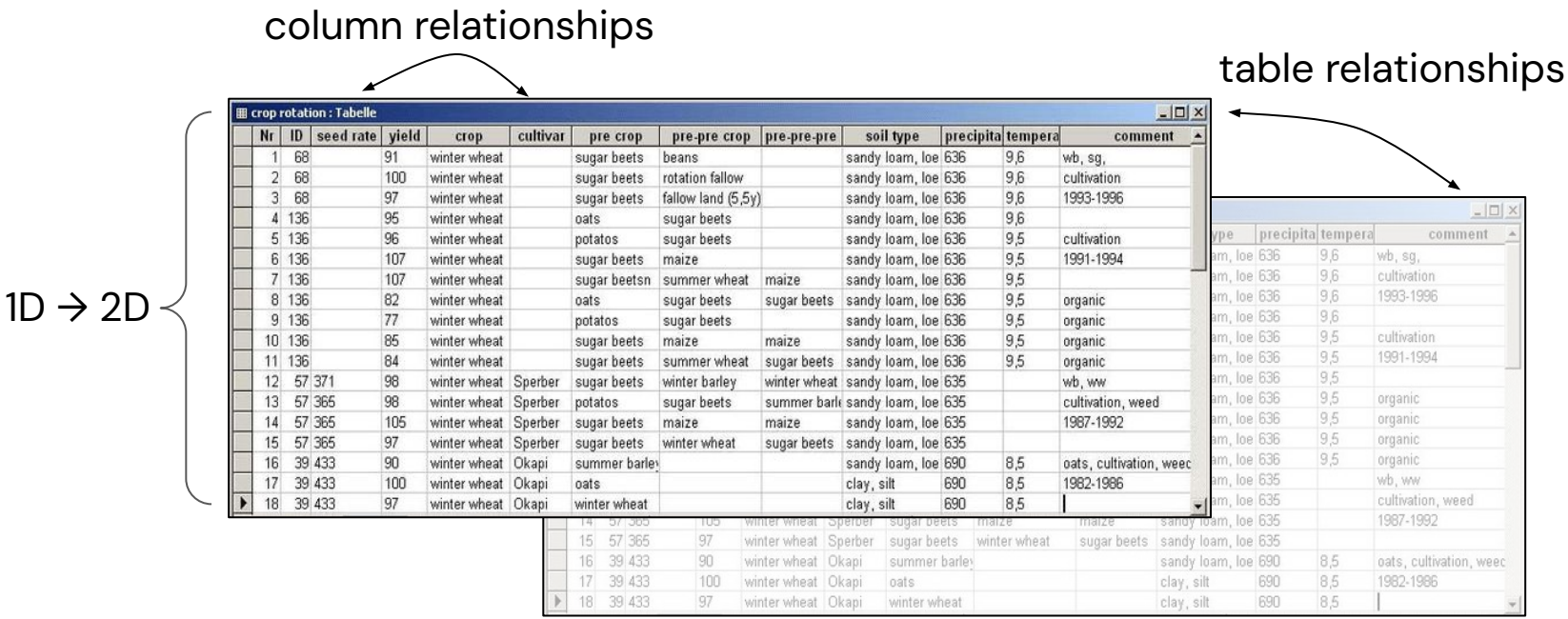
a lot of data to learn!

"Position: Unlocking the Full Potential of Data Science Requires Tabular Foundation Models, Agents, and Humans" by Cong, et al., 2025

The **Challenges** of Tabular AI 💛



Challenge 1: Structure



Nr	ID	seed rate	yield	crop	cultivar	pre crop	pre-pre crop	pre-pre-pre	soil type	precipita	tempera	comment
1	68		91	winter wheat		sugar beets	beans		sandy loam, loe	636	9,6	wb, sg,
2	68		100	winter wheat		sugar beets	rotation fallow		sandy loam, loe	636	9,6	cultivation
3	68		97	winter wheat		sugar beets	fallow land (5,5y)		sandy loam, loe	636	9,6	1993-1996
4	136		95	winter wheat		oats	sugar beets		sandy loam, loe	636	9,6	
5	136		96	winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	cultivation
6	136		107	winter wheat		sugar beets	maize		sandy loam, loe	636	9,5	1991-1994
7	136		107	winter wheat		sugar beetsn	summer wheat	maize	sandy loam, loe	636	9,5	
8	136		82	winter wheat		oats	sugar beets	sugar beets	sandy loam, loe	636	9,5	organic
9	136		77	winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	organic
10	136		85	winter wheat		sugar beets	maize	maize	sandy loam, loe	636	9,5	organic
11	136		84	winter wheat		sugar beets	summer wheat	sugar beets	sandy loam, loe	636	9,5	organic
12	57	371	98	winter wheat	Sperber	sugar beets	winter barley	winter wheat	sandy loam, loe	635		wb, ww
13	57	365	98	winter wheat	Sperber	potatos	sugar beets	summer barl	sandy loam, loe	635		cultivation, weed
14	57	365	105	winter wheat	Sperber	sugar beets	maize	maize	sandy loam, loe	635		1987-1992
15	57	365	97	winter wheat	Sperber	sugar beets	winter wheat	sugar beets	sandy loam, loe	635		
16	39	433	90	winter wheat	Okapi	summer barley			sandy loam, loe	690	8,5	oats, cultivation, weec
17	39	433	100	winter wheat	Okapi	oats			clay, silt	690	8,5	1982-1986
18	39	433	97	winter wheat	Okapi	winter wheat			clay, silt	690	8,5	

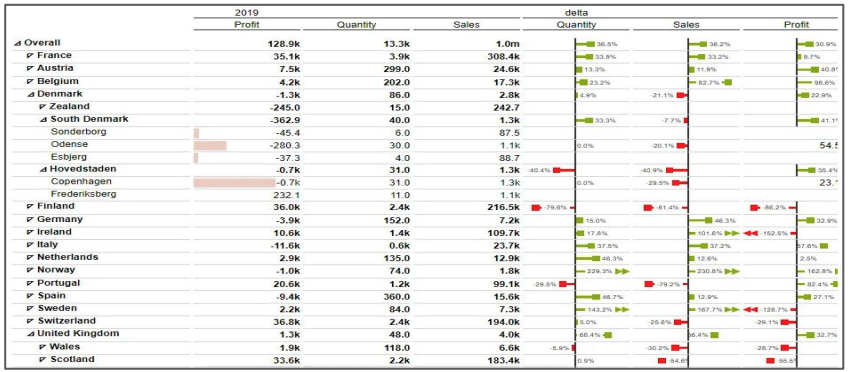
Cong, Hulsebos, et al., Observatory: Characterizing Relational Table Embeddings. *VLDB, 2023.*
Van Breugel & Van der Schaar, Position: Why Tabular Foundation Models Should Be a Research Priority. ICML, 2024.

Challenge 2: Data Heterogeneity

Tables come w/ mixed **data types**, **hierarchies**, **semantics** and **sizes**…

crop rotation : Tabelle

Nr	ID	seed rate	yield	crop	cultivar	pre crop	pre-pre crop	pre-pre-pre	soil type	precipita	tempera	comment
1	68		91	winter wheat		sugar beets	beans		sandy loam, loe	636	9,6	wb, sg,
2	68		100	winter wheat		sugar beets	rotation fallow		sandy loam, loe	636	9,6	cultivation
3	68		97	winter wheat		sugar beets	fallow land (5,5y)		sandy loam, loe	636	9,6	1993-1996
4	136		95	winter wheat		oats	sugar beets		sandy loam, loe	636	9,6	
5	136		96	winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	cultivation
6	136		107	winter wheat		sugar beets	maize		sandy loam, loe	636	9,5	1991-1994
7	136		107	winter wheat		sugar beetsn	summer wheat	maize	sandy loam, loe	636	9,5	
8	136		82	winter wheat		oats	sugar beets	sugar beets	sandy loam, loe	636	9,5	organic
9	136		77	winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	organic
10	136		85	winter wheat		sugar beets	maize	maize	sandy loam, loe	636	9,5	organic
11	136		84	winter wheat		sugar beets	summer wheat	sugar beets	sandy loam, loe	636	9,5	organic
12	57	371	98	winter wheat	Sperber	sugar beets	winter barley	winter wheat	sandy loam, loe	635		wb, ww
13	57	365	98	winter wheat	Sperber	potatos	sugar beets	summer barl	sandy loam, loe	635		cultivation, weed
14	57	365	105	winter wheat	Sperber	sugar beets	maize	maize	sandy loam, loe	635		1987-1992
15	57	365	97	winter wheat	Sperber	sugar beets	winter wheat	sugar beets	sandy loam, loe	635		
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Cong, Hulsebos, et al., Observatory: Characterizing Relational Table Embeddings. *VLDB, 2023.*
Van Breugel & Van der Schaar, Position: Why Tabular Foundation Models Should Be a Research Priority. ICML, 2024.

The **task**



Tabular Reasoning

Today: **different flavors hence needs for tabular reasoning**

Scope: **semantic**, **analytical**, and **inferential** **reasoning**

Back to the Basics: *Semantic* Reasoning

What is this table about?



crop rotation : Tabelle

Nr	ID	seed rate	yield	crop	cultivar	pre crop	pre-pre crop	pre-pre-pre	soil type	precipita	tempera	comment
1	68		91	winter wheat		sugar beets	beans		sandy loam, loe	636	9,6	wb, sg,
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5	136		96	winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	cultivation
6	136		107	winter wheat		sugar beets	maize		sandy loam, loe	636	9,5	1991-1994
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8	136		82	winter wheat		oats	sugar beets	sugar beets	sandy loam, loe	636	9,5	organic
9	136		77	winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	organic
10	136		85	winter wheat		sugar beets	maize	maize	sandy loam, loe	636	9,5	organic
11	136		84	winter wheat		sugar beets	summer wheat	sugar beets	sandy loam, loe	636	9,5	organic
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13	57	365	98	winter wheat	Sperbe	potatos	sugar beets	summer barl	sandy loam, loe	635		cultivation, weed
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18	39	433	97	winter wheat	Okapi	winter wheat			clay, silt	690	8,5	

Pre crop?
Pre-pre crop?
Pre-pre-pre?

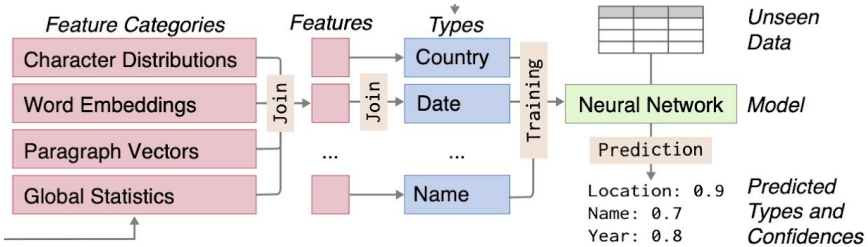
Semantics key in downstream tasks (data cleaning, integration, predictive ML)

Sherlock: A Deep Learning Approach to Semantic Data Type Detection by Hulsebos et al., KDD, 2019

Can we infer *column semantics* w/ embeddings?

Sherlock: neural model w/ thousands of params trained on column representations

78 semantic types (`name`, `address`, etc)



Performance (F1)

Sherlock	0.89
Human Consensus	0.32

🐘 **LLMs?** Sherlock (*tiny* model) outperformed (*huge*) GPT-3.5 with +6%

Sherlock: A Deep Learning Approach to Semantic Data Type Detection by Hulsebos et al., KDD, 2019

Utility: from types to **sensitivity**?

Detecting Personal Information in Training Corpora: an Analysis

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Jesse Dodge
AI2, USA

Margaret Mitchell
Hugging Face, USA

Abstract

Large language models are trained on increasing quantities of unstructured text, the largest sources of which are scraped from the Web. These Web scrapes are mainly composed of heterogeneous collections of text from multiple domains with minimal documentation. While some work has been done to identify and remove toxic, biased, or sexual language, the topic of *personal information* (PI) in textual data used for training Natural Language Processing (NLP) models is relatively underexplored. In this work, we draw from definitions of PI across multiple countries to define the first PI taxonomy of its kind, categorized by type and risk level. We then conduct a case study on the Colossal Clean Crawled Corpus (C4) and the Pile, to detect some of the highest-risk personal information, such as email addresses and credit card numbers, and examine the differences between automatic and

pretrained models to generate arbitrary sequences without modifying the underlying weights at all via steering vectors (Subramani et al., 2019; Subramani and Suresh, 2020; Subramani et al., 2022) and prompting (Shin et al., 2020; Li and Liang, 2021).

This suggests that it is necessary to better understand the types of PI contained in training corpora and the types of harms that they can cause, and to propose ways for automatically detecting (and, eventually, removing) the most high-risk types of PI from NLP corpora. We endeavor to address both of these directions in the current article: we start with defining different types of PI and propose a novel categorization in Section 2 and discuss the risks of different types of PI. Then, in Section 3, we explore the difficulty in detecting one of the highest-risk and easiest-to-identify types of PI, CHARACTER-BASED identifiers, comparing a

We argue:

data sensitivity goes beyond types

"Data whose sensitivity depends on external factors and requires protection due to how, by whom, and in what context it can be misused."

Towards Contextual Sensitive Data Detection, Telkamp & Hulsebos, Findings of EMNLP, 2026.

Data sensitivity is contextual!

🤔 Q1: are **all** PI typed columns, really sensitive?

personal `address` of human aid workers

Name	Role	Address	Region
Lina Haddad	Field Coordinator	14 Olive Grove Rd, Sector 3, Al Noor District	Northern Syria
David Mensah	Medical Logistics Officer	House 8, Kivu Relief Compound, East Ridge	Eastern DRC
Sara Petrov	Emergency Specialist	22 Cedar Lane, Block B, River Quarter	South Sudan

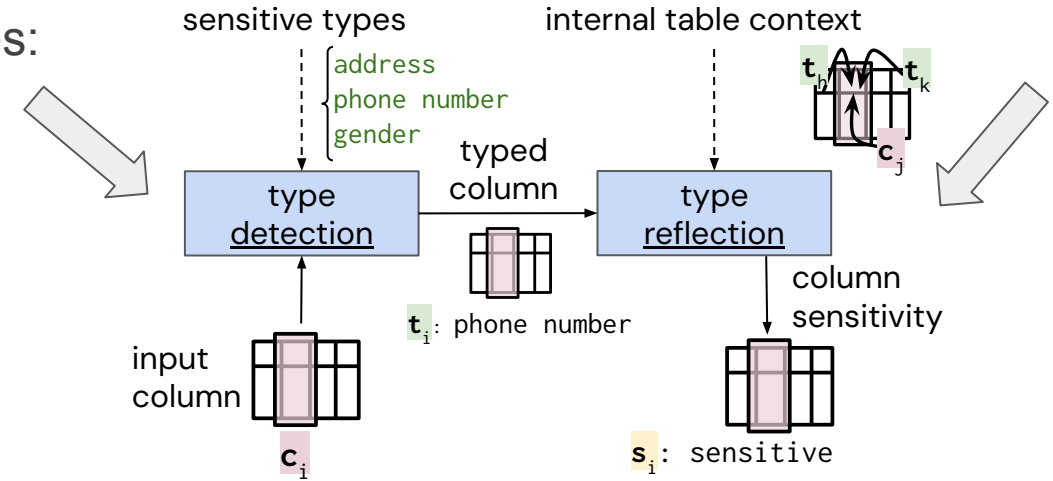
organizational `address` of professors

Name	Department	Address	Institution
Dr. Elena Vos	Sociology	Room 4.12, Social Sciences Building, 120 University Ave	Utrecht University
Prof. Michael Chen	Physics	Office 2-318, Science Hall, 85 Academic Way	TU Delft
Dr. Amina Rahman	History	Room 5B, Humanities Centre, 44 College Street	Leiden University

Towards Contextual Sensitive Data Detection, Telkamp & Hulsebos, Findings of EMNLP, 2026.

Mechanism: *type-contextualization*

w/o PII types:
recall -5%



w/o reflection:
precision -16%

✨**integrated** in UN's humanitarian data platform

🏆won Amsterdam AI Thesis Award

94% sensitive PII detected
vs 63% with Google's DLP!

*Experiments with **Qwen3 14B**

Towards Contextual Sensitive Data Detection, Telkamp & Hulsebos, Findings of EMNLP, 2026.

Context beyond the table matters, too

🤔 Q2: are **only** PI typed columns, sensitive?

`geo-coordinates` from hospitals in **Gaza**

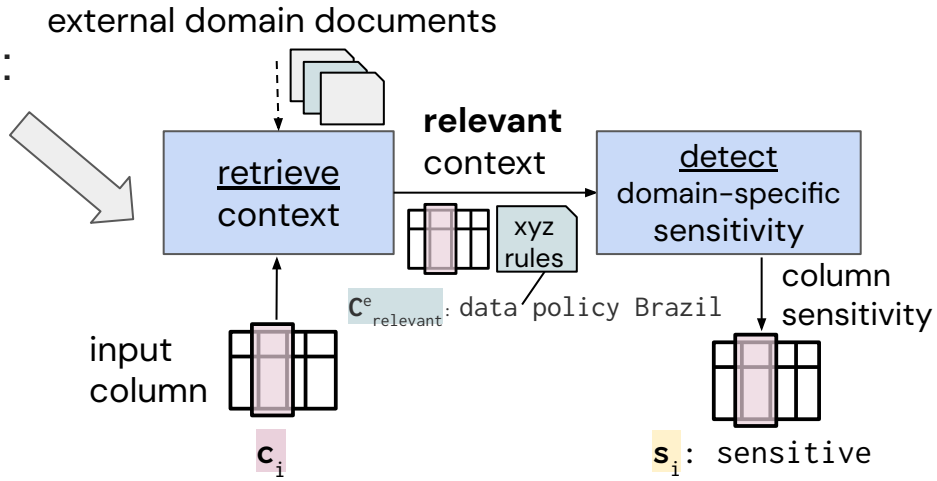
Facility Name	Lat	Long	Region
Al Amal Field Hospital	31.4872	34.4386	Central Gaza
Coastal Emergency Clinic	31.5129	34.4551	Northern Gaza
Rafah Community Hospital	31.2964	34.2478	Southern Gaza

`geo-coordinates` from hospitals in **Germany**

Facility Name	Lat	Long	Region
St. Marien Medical Center	52.5218	13.4097	Berlin
Rhine General Hospital	50.9386	6.9603	Cologne
Bavaria University Hospital	48.1397	11.5654	Munich

Mechanism: *domain-contextualization*

w/o policy rules:
precision -8%



✨**integrated** in UN's humanitarian data platform

🏆won Amsterdam AI Thesis Award

100% sensitive non-PII data detected but still yields some FPs

*Experiments with **Qwen3 14B**

Towards Contextual Sensitive Data Detection, Telkamp & Hulsebos, Findings of EMNLP, 2026.

Domain context: UN data policies

INFORMATION SHARING PROTOCOL
The Occupied Palestinian Territory
25 April 2025

OVERVIEW
This Information Sharing Protocol (ISP) is designed to support *data responsibility -- the safe, ethical, and effective management of data* - within the Occupied Palestinian Territory (The OPT). It establishes a collective framework for information and data exchange on the basis of a common data and information sensitivity classification.

The OPT Assessment and Information Management Working Group (AIMWG) developed this ISP through a collective exercise in accordance with the Inter-Agency Standing Committee (IASC) Operational Guidance on Data Responsibility.[1] The ISP was endorsed by the OPT Inter-Cluster Coordination Group (ICCG) in June 2024 and revised by the AIMWG and ICCG the 26 and 28 February 2025

In this context, this ISP serves as the primary document governing data and information sharing in the humanitarian response in The OPT. This ISP should be used as a reference in the exchange of data and information with the Government of The OPT. This ISP complements existing policies and guidelines and does not in any way affect or replace obligations and restrictions contained in applicable legal and regulatory frameworks, Cluster-

High Sensitivity Information or data that, if disclosed or accessed without proper authorization, **are likely to cause serious harm or negative impacts** to affected people and/or humanitarian actors and/or damage to a response.	• **Needs assessment** data at the neighborhood (ADM4) level or lower • **Survey results** at the neighborhood (ADM4) level or lower • **Explosive ordnance contamination** data (exact locations or at ADM3 level) • **Aid-worker contact details** and contact lists if consent for sharing has not been obtained • **Minutes** of internal meetings • **Information on evacuation of affected people** or where local organizations are supporting relocation	**Classification: Confidential** Data or information can be disclosed within an organization or small community of organizations directly involved in delivering humanitarian assistance, based on a clearly specified purpose and related standards for data responsibility, including data protection. Methods for sharing confidential data: • Microsoft Teams files, SharePoint, OneDrive and similar collaboration platforms • Internal intra-Cluster sharing • Inter-Cluster sharing on case-by-case basis • HDX via HDX Connect • ArcGIS Online
Severe Sensitivity Information or da… that, if disclosed … accessed without proper authorization, **are likely to cause severe harm or negative impacts and/or damage** to affected people and/or humanitarian actors and/or impede	• **Humanitarian notification information** • **Raw survey** data, such as individual survey responses, at the respondent or household level • **Data on attacks on health facilities** with locations (e.g. coordinates) • **5W** data at the neighborhood (ADM4) level or site (ADM5) level • Live data on population movement based on **metadata** generated by electronic devices or based on	**Classification: Strictly Confidential** Highly limited, bilateral disclosure only. Determined and approved on a case-by-case basis, with assurance of upholding the highest standards of data responsibility, including data protection. Methods for sharing strictly confidential data:

https://www.unocha.org/publications/report/occupied-palestinian-territory/information-sharing-protocol-occupied-palestinian-territory-april-2025

UN example case

Reported location on NFIs suppliers							
nfi_supplier_	nfi_supplier_loc- value -from_Iran	n	i_supplier_	nfi_supplier_	nfi_supplier_	nfi_supplier_	nfi_supplier_	nfi_supplier_
NA	0%		0%	0%	100%	0%	0%	0%
NA	0%		0%	0%	33%	0%	67%	0%
NA	0%		0%	0%	100%	0%	0%	0%

1. interpretation of data

2. assessment of sensitivity

3. context-grounded explanation

Context-grounded LLM output

1. The column “nfi_supplier_loc- value -from_Iran” indicates the origin of NFIs suppliers.

2. This information, even with country-level disaggregation, doesn’t reveal personal or sensitive data. It relates to the location of suppliers, which is generally not considered sensitive.

3. It aligns with the ISP’s guideline on “Facility data (e.g., health, education, water points) at national or regional level, unless explicitly restricted by clusters.” Since this column is about locations of suppliers and not about individuals or specific operations, it is categorized as non-sensitive.

✨rule-grounding has **utility beyond sensitivity**, eg reviewing insurance, loan, mortgage, damage requests

Towards Contextual Sensitive Data Detection, Telkamp & Hulsebos, Findings of EMNLP, 2026.

From *Semantic* to *Analytical* Reasoning

Can we **also just use LLMs** for analytical querying?

*What is the **average budget** of Drama movies with a **greater or equal to 8.0** rating?*

MovieID	Title	MPAA Rating	Budget	Gross	Release Date	Genre	Runtime	Rating	Rating Count
121	Dangerous Minds	R	23 000 000	179 519 401	1995-08-11	Crime	99	6.5	45 479
377	Slumdog Millionaire	R	15 000 000	377 910 544	2008-05-12	Drama	120	8.0	758 215
386	Kung Fu Panda	PG	130 000 000	631 744 560	2008-06-04	Animation	90	7.5	390 478

ActorID	Name	Year of birth	Birth City	Birth Country	Height	Biography	Gender	Ethnicity	NetWorth
410	Dustin Hoffman	1937-08-08	LA	USA	65.0	“”	Male	-	$100M
553	George Dzundza	1945-07-19	Rosenheim	Germany	69.0	“”	Male	White	$2M
2099	Saurabh Shukla	1963-03-05	-	India	-	“”	-	-	-

MovieID	ActorID	Character Name	creditOrder
121	553	Hal Griffith	2
377	2099	Sergeant Srinivas	2
386	410	Shifu	2

DeepSeek reasoning trace:

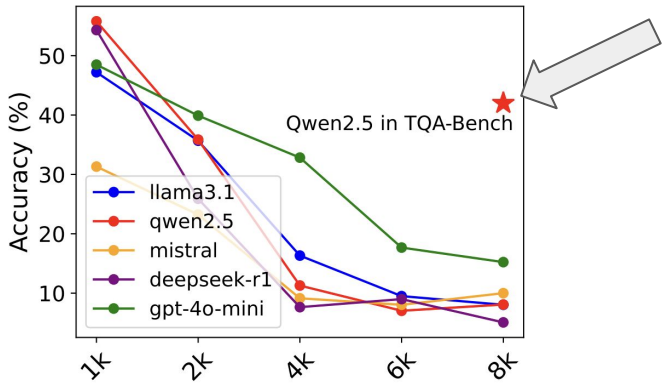
“... Wait no—the data doesn’t show that...
... Wait I’m getting confused...”

How well do LLMs reason over tabular data, really? by Wolff & Hulsebos, TRL@ACL, 2025

CWI

The expected sad state of tabular QA w/ LLMs

LLM-as-a-judge* on *averaging* questions over multiple tables in TQA-Bench



MPC eval is a bad idea

*>0.95% llm-as-a-judge accuracy (only 0.8% incorrect answers are judged as correct)

How well do LLMs reason over tabular data, really? by Wolff & Hulsebos, TRL@ACL, 2025

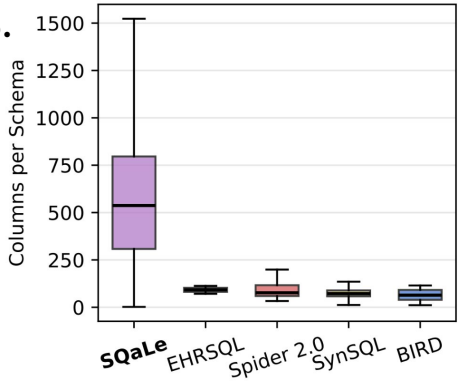
Better use SQL… (through LLM code generation)

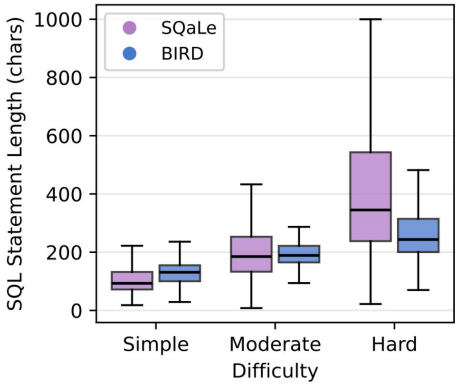
Text-to-SQL is not solved:

- SOTA: throwing huge LLM pipelines at it? Not scalable for “BI agentification”.
- Complex queries/schemas: text-to-SQL needs *data-grounding*.

SQaLe dataset to specialize models.

- First **large** text-to-SQL dataset
- 700K+ (question, sql, schema)
- Queries over 9K+ **real** schemas from SchemaPile: large & diverse.





SQALE: A Large Dataset for Specialized Text-to-SQL by Wolff, Gomm, and Hulsebos, in review, 2026

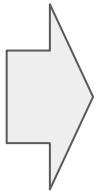
CWI

Realistic queries in SQaLe?

An example question in **SQaLe**:

"For each customer in the gold or platinum loyalty tier who has placed at least two delivered orders in the last six months, return their full name, email, city and state, the total number of delivered orders, the total amount they paid based only on captured payments, their most frequently ordered product category in that period, and whether any of the products they ordered currently have low inventory in any warehouse (defined as quantity on hand minus quantity reserved less than 10), sorted by total captured payment amount from highest to lowest."

*SQaLe has different verbosity levels :)



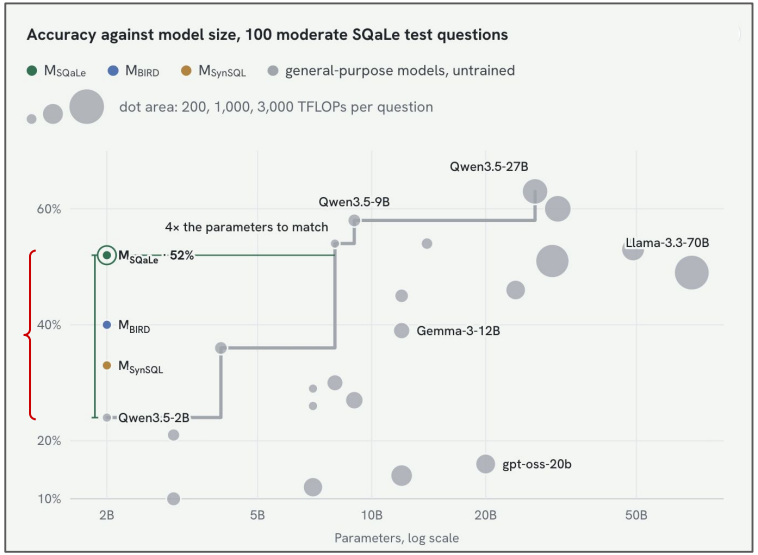
```
WITH gold_platinum_customers AS
  (SELECT c.id,
          c.first_name,
          c.last_name,
          c.email,
          c.city,
          c.state
   FROM customers c
   JOIN orders o ON c.id = o.customer_id
   WHERE c.loyalty_tier IN ('gold',
                            'platinum')
     AND o.status = 'delivered'
     AND o.delivered_at >= date('now', '-6 months')
   GROUP BY c.id
   HAVING COUNT(*) >= 2),
     order_details AS
  (SELECT gpc.id,
          gpc.first_name,
          gpc.last_name,
          gpc.email,
          gpc.city,
          gpc.state,
          o.order_number,
          SUM(p.amount) AS total_paid,
          p.status AS payment_status
   FROM gold_platinum_customers gpc
   JOIN orders o ON gpc.id = o.customer_id
   LEFT JOIN payments p ON o.id = p.order_id
   AND p.status = 'captured'
   WHERE o.delivered_at >= date('now', '-6 months')
     AND o.status IN ('delivered',
                      'completed')
   GROUP BY gpc.id,
            o.id),
     product_category AS
  (SELECT od.id,
          od.first_name,
          od.last_name,
          od.email,
          od.city,
          od.state,
          p.category,
          ROW_NUMBER() OVER (PARTITION BY od.id
                             ORDER BY COUNT(*) DESC) AS rn
   FROM order_details od
   JOIN order_items oi ON od.order_number = CAST(oi.order_id AS TEXT)
   JOIN products p ON oi.product_id = p.id
   GROUP BY od.id)
SELECT gpc.first_name,
       gpc.last_name,
       gpc.email,
       gpc.city,
       gpc.state,
       COUNT(od.order_number) AS total_orders,
       SUM(od.total_paid) AS total_captured_payment_amount
FROM gold_platinum_customers gpc
JOIN order_details od ON gpc.id = od.id
AND od.payment_status = 'captured'
LEFT JOIN product_category pc ON od.id = pc.id
AND pc.rn = 1
WHERE pc.category IS NOT NULL
GROUP BY gpc.id
ORDER BY total_captured_payment_amount DESC;
```

SQALE: A Large Dataset for Specialized Text-to-SQL by Wolff, Gomm, and Hulsebos, in review, 2026

CWI

SQaLe enables our vision of *specialized small* text-to-SQL

Qwen-2B trained for agentic deployment using RL



Training on SQaLe much **better than** training on **equally large synthetic** corpus or **a few real DBs**.

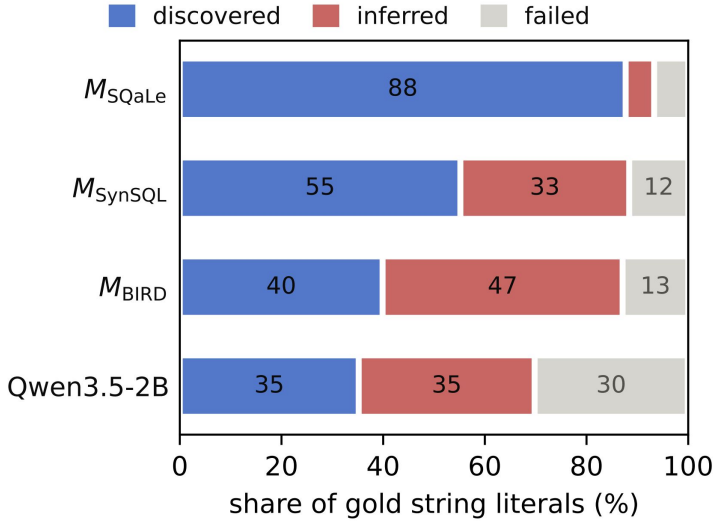
Why?

SQALE: A Large Dataset for Specialized Text-to-SQL by Wolff, Gomm, and Hulsebos, in review, 2026

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Agents *need to learn to* **explore the data**!

By training on **large populated** schemas, ***data exploration* skills *emerge*** in the agentic model – while other models copy values from the question.

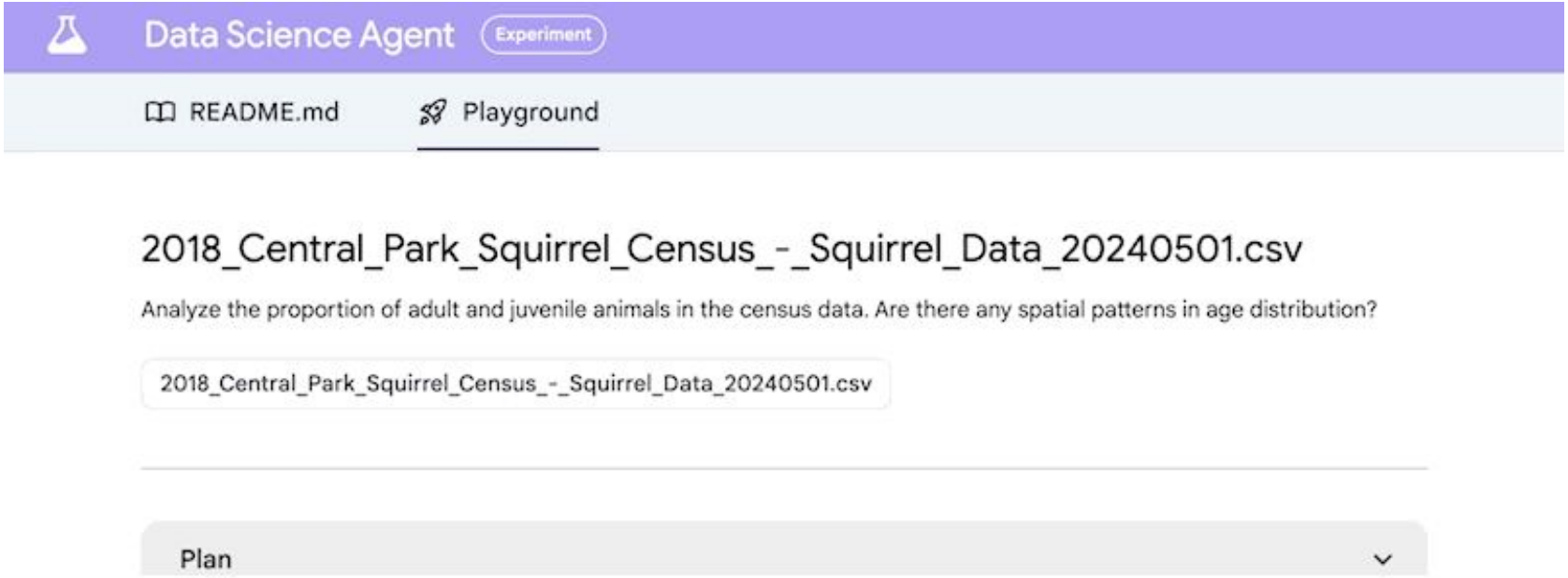


SQALE: A Large Dataset for Specialized Text-to-SQL by Wolff, Gomm, and Hulsebos, in review, 2026

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From *Analytical* to *Inferential* Reasoning

Flashy demos of Data Science Agents



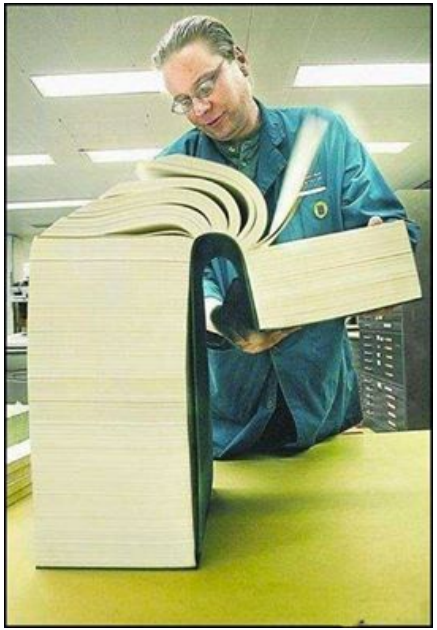
Problem Solved?

Model	*SDE* (69 tasks) Success	*SDE* (69 tasks) Score	*PM* (28 tasks) Success	*PM* (28 tasks) Score	*DS* (14 tasks) Success	*DS* (14 tasks) Score
Closed model APIs						
Claude-3.5-Sonnet	30.43	38.02	35.71	51.31	14.29	21.70
Gemini-2.0-Flash	13.04	18.99	17.86	31.71	0.00	6.49
GPT-4o	13.04	19.18	17.86	32.27	0.00	4.70
Gemini-1.5-Pro	4.35	5.64	3.57	13.19	0.00	4.82
Amazon-Nova-Pro-v1	2.90	6.07	3.57	12.54	0.00	3.27
Open-weight models						
Llama-3.1-405b	5.80	11.33	21.43	35.62	0.00	5.42
Llama-3.3-70b	11.59	16.49	7.14	19.83	0.00	4.70
Qwen-2.5-72b	7.25	11.99	10.71	22.90	0.00	5.42
Llama-3.1-70b	1.45	4.77	3.57	15.16	0.00	5.42
Qwen-2-72b	2.90	3.68	0.00	7.44	0.00	4.70

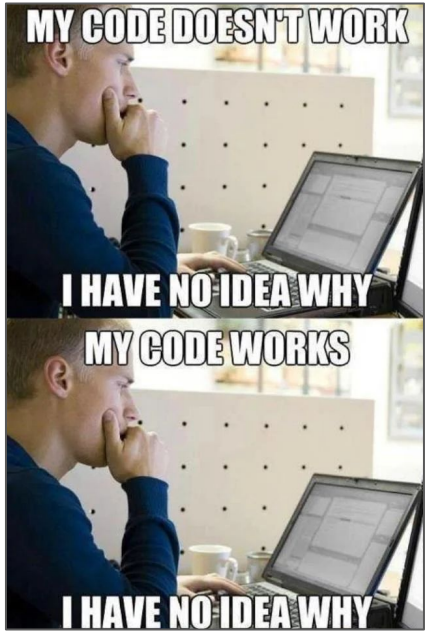
From *TheAgentCompany: Benchmarking LLM Agents on Consequential Real World Tasks* by Xu et al., NeurIPS, 2025.

How do we measure success?

Human evaluation….?



Automated evaluation…?



Data science is…

- Not just code generation!
- Not just getting a reasonable accuracy!

→ **Methodological failures creep in silently!**

The Silent Failures in Agentic Data Science by Li, X., et al, in review, 2026.

The Silent Failures in Agentic Data Science

Analyzed failure modes in **220 DS tasks:**
- 3 data science agents
- 3 different models

(validated) LLM-judge

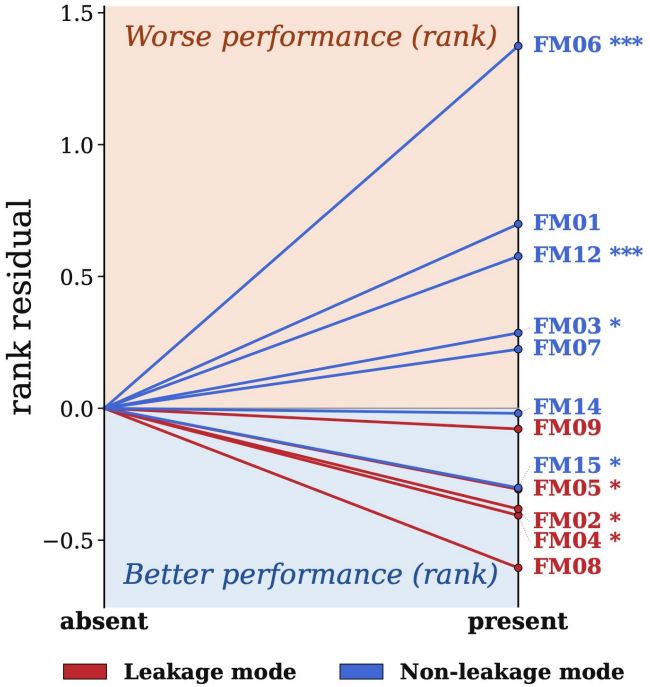
Agent	Model	Wrong task formulation	Peeking at test data during EDA	Not cleaning data	Imputation before splitting	Ignoring temporal order	No train/val split	No feature engineering	Scaling before splitting	Feature selection before splitting	Wrong model type for task	No regularisation in DL	Naive ensembling	Reusing test set across variants	Single eval of stochastic model	Using only one eval metric
ReAct	Qwen-4B	0.02	0.01	0.28	0.00	0.00	1.00	0.98	0.00	0.00	0.01	N/A	N/A	N/A	0.48	0.15
	Qwen-27B	0.01	0.12	0.04	0.13	0.36	0.05	0.50	0.00	0.03	0.00	N/A	0.49	0.00	0.95	0.89
	Gemini-3.7	0.02	0.07	0.30	0.06	0.48	0.40	0.38	0.00	0.08	0.00	N/A	0.90	N/A	0.04	0.14
AIDE	Qwen-4B	0.01	0.02	0.72	0.08	0.13	0.07	0.93	0.02	0.00	0.03	N/A	N/A	N/A	0.19	0.87
	Qwen-27B	0.03	0.08	0.21	0.03	0.15	0.01	0.71	0.00	0.02	0.00	N/A	0.81	N/A	0.08	0.85
	Gemini-3.7	0.01	0.15	0.45	0.13	0.47	0.00	0.12	0.03	0.14	0.00	N/A	0.46	0.00	0.03	0.88
MLEvolve	Qwen-4B	0.02	0.02	0.86	0.09	0.02	0.69	0.94	0.02	0.00	0.00	N/A	N/A	N/A	1.00	0.34
	Qwen-27B	0.01	0.08	0.17	0.03	0.17	0.01	0.16	0.00	0.01	0.00	0.00	0.51	0.06	0.76	0.87
	Gemini-3.7	0.00	0.10	0.10	0.10	0.36	0.00	0.00	0.01	0.06	0.00	0.00	0.04	0.01	0.13	0.43

Besides common DS FMs, we found **many flavors of data leakage**.

The Silent Failures in Agentic Data Science by Li, X., et al, in review, 2026.

CWI

Stop relying only downstream performance!



"cheating" models

↓

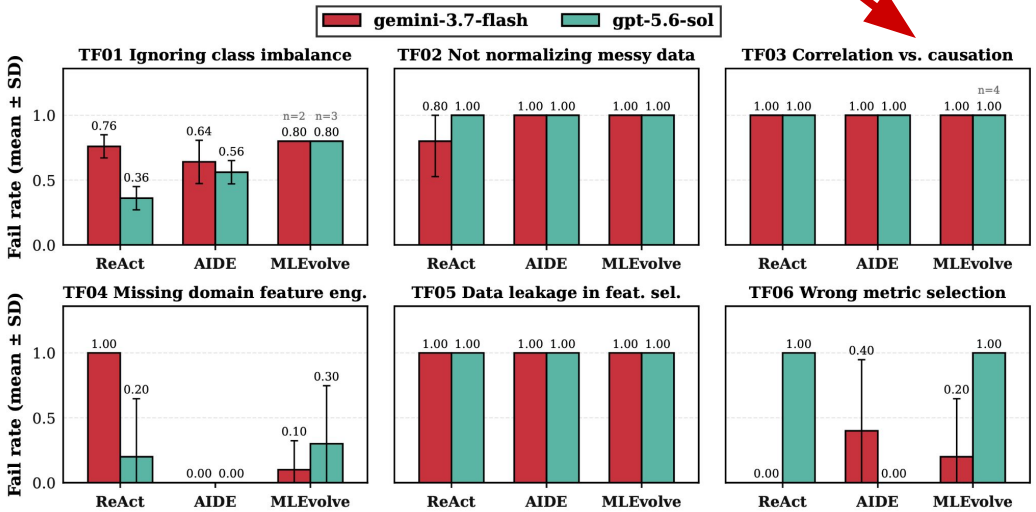
better performance

The Silent Failures in Agentic Data Science by Li, X., et al, in review, 2026.

Maybe, better models do better?

Also developed a **test suite with a few controlled tests.**

e.g., adding correlating “proxy” of target



✨ **Hongqian Xia, who’s now at Booking** helped develop this test suite!

The Silent Failures in Agentic Data Science by Li, X., et al, in review, 2026.

Future of (Agentic) Data Science…

LLMing won't do it, code generation won't do it either…

→ data science agents need good data science practices

Some **opportunities**:

- Capturing specific "skills", building rigid harnesses, models?
- Plug in tabular foundation models
- Train human experts to focus more on, e.g., problem formulation, data selection, specialized feature engineering, etc!

Recap on the Flavors of Tabular Reasoning

Wrapping up…

- ➔ **Tabular AI is Exciting!!!**
- ➔ For tasks powered by table semantics (e.g. sensitivity val), **domain-contextualization** is key for LLMs to improve, and support interpretation and consistency of decisions.
- ➔ Tabular AI is **multifaceted**, no future for "general-purpose (tabular) model": **semantic / analytical / statistical reasoning are different in nature**. Future for ***small specialized*** models, text2sql, retrieval, prediction…
- ➔ Needs of *agentic data science* for predictive tasks go beyond contextualized LLMs/code gen: domain- & data science **knowledge** are key. Human data science experts can now really move the needle.

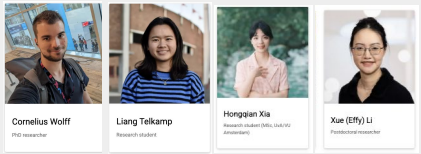
That’s it!

TRL Lab

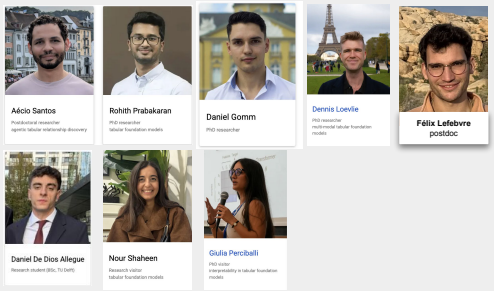
trl-lab.github.io

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Leads of projects



Other members



Funding & Collaboration

CWI Centrum Wiskunde & Informatica
ellis UNIT AMSTERDAM
MIT
Berkeley UNIVERSITY OF CALIFORNIA
Universiteit van Amsterdam
SAP
NWO
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